

Structural Organisation in Animals

In the animal kingdom you meet a large variety of organisms, both unicellular and multicellular. In **unicellular organisms**, all functions like **digestion, respiration and reproduction** are performed by a **single cell**. In the complex body of **multicellular animals** the same basic functions are carried out by **different groups of cells** in a well organised manner.

The body of a simple organism like **Hydra** is made of different types of cells, and the number of cells in each type can be in **thousands**. The **human body** is composed of **billions of cells** to perform various functions.

● **Tissue** - in multicellular animals, a **group of similar cells** along with **intercellular substances** that together perform a **specific function** (one or more functions) in the body.

You may be surprised to know that **all complex animals consist of only four basic types of tissues**. These tissues are organised in specific proportion and pattern to form an **organ** like stomach, lung, heart and kidney. When **two or more organs** perform a common function by their physical and/or chemical interaction, they together form an **organ system**, e.g., digestive system, respiratory system, etc.

Cells, tissues, organs and organ systems split up the work in a way that exhibits **division of labour** and contribute to the **survival of the body as a whole**.

7.1

Animal Tissues

The **structure of the cells vary according to their function**. Therefore, the tissues are different and are broadly classified into **four types**: (i) **Epithelial**, (ii) **Connective**, (iii) **Muscular** and (iv) **Neural**.

7.1.1

Epithelial Tissue

We commonly refer to an epithelial tissue as **epithelium** (pl.: epithelia). This tissue has a **free surface**, which faces either a body fluid or the outside environment and thus provides a **covering or a lining** for some part of the body. Epithelia are **sheet-like tissues** lining the body's surface and its **cavities, ducts and tubes**.

The cells are **compactly packed with little intercellular matrix**. There are **two types** of epithelial tissues namely **simple epithelium** and **compound epithelium**. **Simple epithelium** is composed of a **single layer of cells** and functions as a lining for body cavities, ducts and tubes. The **compound epithelium** consists of **two or more cell layers** and has a **protective function** as it does in our skin.

On the basis of structural modification of the cells, **simple epithelium** is further divided into **three types** (Figure 7.1): (i) **Squamous**, (ii) **Cuboidal**, (iii) **Columnar**.

Simple epithelium	Structure	Location	Main function
Squamous	Single thin layer of flattened cells with irregular boundaries	Walls of blood vessels and air sacs of lungs	Forms a diffusion boundary
Cuboidal	Single layer of cube-like cells	Ducts of glands and tubular parts of nephrons in kidneys	Secretion and absorption
Columnar	Single layer of tall and slender cells; nuclei at the base; free surface may have microvilli	Lining of stomach and intestine	Secretion and absorption

The epithelium of the **proximal convoluted tubule (PCT)** of the nephron in the kidney has **microvilli**.

- If the columnar or cuboidal cells bear **cilia** on their free surface they are called **ciliated epithelium** (Figure 7.1d). Their function is to **move particles or mucus in a specific direction** over the epithelium. They are mainly present in the inner surface of **hollow organs like bronchioles and fallopian tubes**.
- Some of the columnar or cuboidal cells get specialised for secretion and are called **glandular epithelium** (Figure 7.2). They are mainly of two types: **unicellular**, consisting of isolated glandular cells (**goblet cells** of the alimentary canal), and **multicellular**, consisting of a cluster of cells (**salivary gland**).

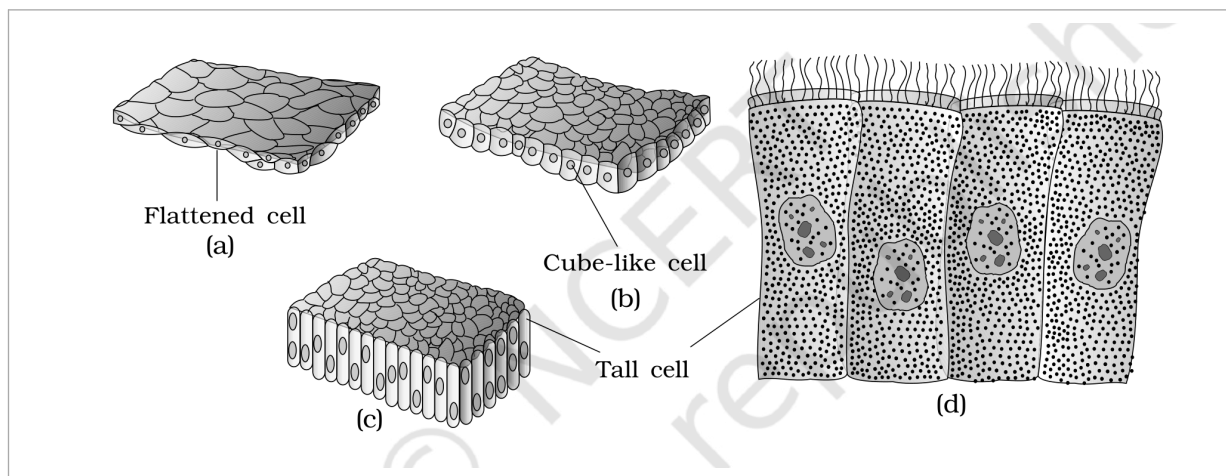


Figure 7.1 Simple epithelium: (a) Squamous (b) Cuboidal (c) Columnar (d) Columnar cells bearing cilia

① [NOTE] Figure 7.1 labels (verbatim): **Flattened cell**; **Cube-like cell**; **Tall cell**.

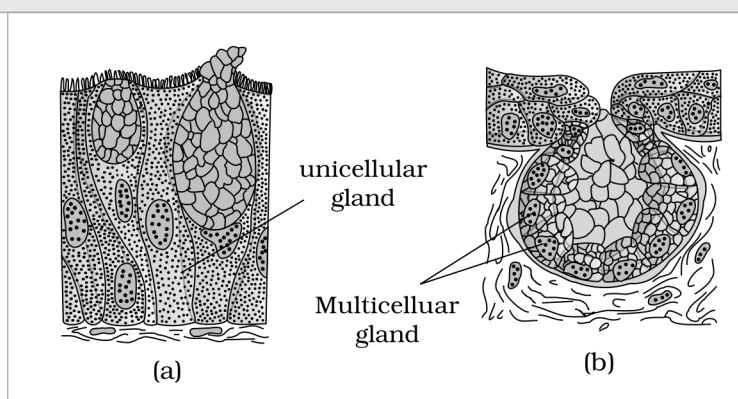


Figure 7.2 Glandular epithelium: (a) Unicellular (b) Multicellular

① [NOTE] Figure 7.2 labels (verbatim): **Unicellular gland**; **Multicellular gland**.

On the basis of the **mode of pouring of their secretions**, glands are divided into two categories namely **exocrine** and **endocrine** glands.

Gland type	Duct	Secretions
Exocrine	Products released through ducts or tubes	Mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products
Endocrine	Do not have ducts	Hormones, secreted directly into the fluid bathing the gland

Compound epithelium is made of **more than one layer** (multi-layered) of cells and thus has a **limited role in secretion and absorption** (Figure 7.3). Their main function is to **provide protection against chemical and mechanical stresses**. They cover the **dry surface of the skin**, the moist surface of **buccal cavity**, **pharynx**, inner lining of ducts of **salivary glands** and of **pancreatic ducts**.

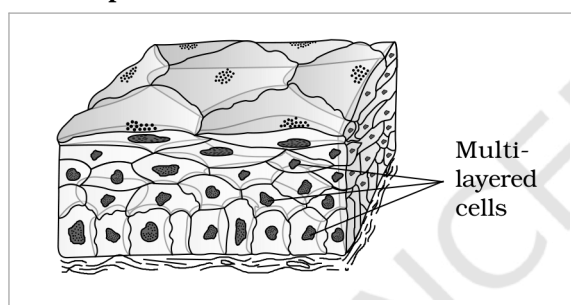


Figure 7.3 Compound epithelium

① [NOTE] Figure 7.3 label (verbatim): **Multi-layered cells**.

All cells in epithelium are held together with **little intercellular material**. In nearly all animal tissues, **specialised junctions** provide both **structural and functional links** between its individual cells. **Three types of cell junctions** are found in the epithelium and other tissues. These are called **tight, adhering and gap junctions**.

Cell junction	Function
Tight junctions	Help to stop substances from leaking across a tissue
Adhering junctions	Perform cementing to keep neighbouring cells together
Gap junctions	Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules

7.1.2 Connective Tissue

Connective tissues are **most abundant and widely distributed** in the body of complex animals. They are named connective tissues because of their special function of **linking and supporting** other tissues/organs of the body. Diverse types of connective tissues **bind together, support, strengthen, protect, and insulate** other tissue in the body. **Connective tissue covers all three** of the other tissue types.

They range from **soft connective tissues** to **specialised types**, which include **cartilage, bone, adipose, and blood**. In all connective tissues **except blood**, the cells secrete **fibres of structural proteins called collagen or elastin**. The fibres provide **strength, elasticity and flexibility** to the tissue. These cells also secrete **modified polysaccharides**, which accumulate between cells and fibres and act as **matrix (ground substance)**. In short, **soft connective tissues** consist of **protein fibres** as well as a variety of **cells arranged in a ground substance**.

Connective tissues are classified into **three types**: (i) **Loose connective tissue**, (ii) **Dense connective tissue** and (iii) **Specialised connective tissue**.

A Loose Connective Tissue

- **Loose connective tissue** has cells and fibres **loosely arranged in a semi-fluid ground substance**, for example, **areolar tissue** present beneath the skin (Figure 7.4). Often it serves as a **support framework for epithelium**. It contains **fibroblasts** (cells that produce and secrete fibres), **macrophages** and **mast cells**.
- **Adipose tissue** is another type of **loose connective tissue** located mainly beneath the skin. The cells of this tissue are **specialised to store fats**. The excess of nutrients which are not used immediately are **converted into fats** and are stored in this tissue. Thus adipose tissue is a **reservoir of stored energy**.

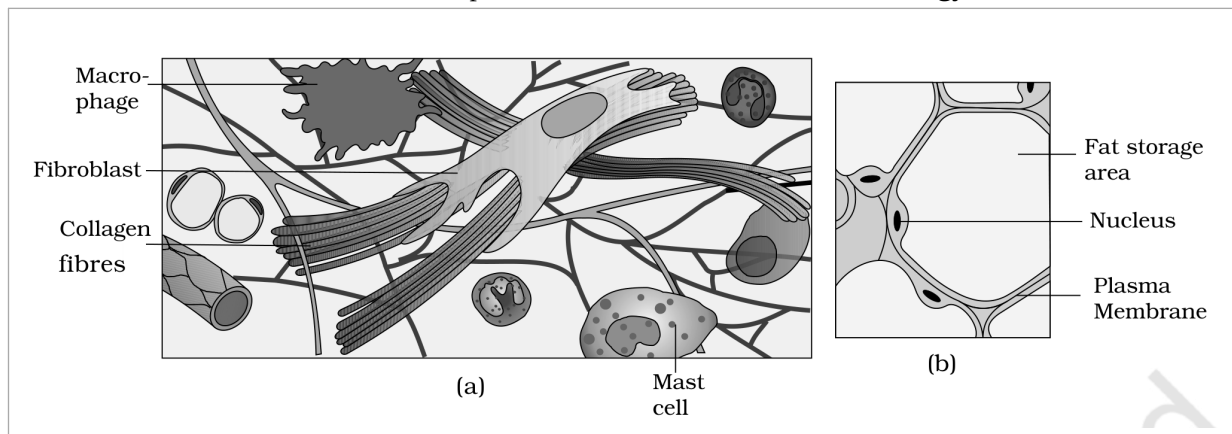


Figure 7.4 Loose connective tissue: (a) Areolar tissue (b) Adipose tissue

① [NOTE] Figure 7.4 labels (verbatim): **Macrophage; Fibroblast; Collagen fibres; Mast cell; Fat storage area; Nucleus; Plasma Membrane.**

B Dense Connective Tissue

Fibres and fibroblasts are **compactly packed** in the dense connective tissues. Orientation of fibres shows a **regular or irregular pattern** and they are called **dense regular** and **dense irregular** tissues.

- In the **dense regular** connective tissues, the **collagen fibres** are present in rows between many parallel bundles of fibres. **Tendons**, which attach skeletal muscles to bones, and **ligaments**, which attach one bone to another, are examples of this tissue.

- **Dense irregular** connective tissue has fibroblasts and many fibres (mostly collagen) that are **oriented differently** (Figure 7.5). This tissue is present in the **skin**.

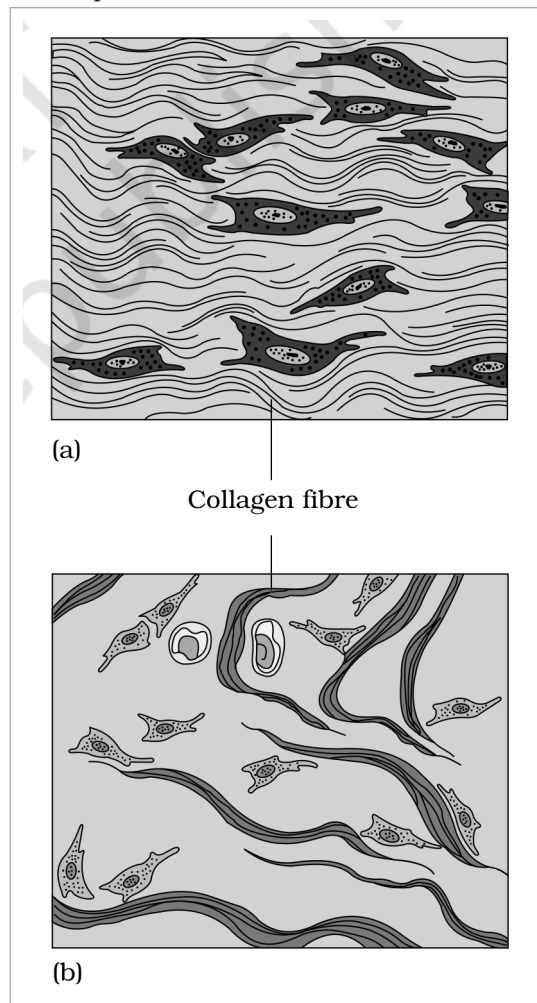


Figure 7.5 Dense connective tissue: (a) Dense regular (b) Dense irregular

① [NOTE] Figure 7.5 label (verbatim): **Collagen fibre**.

c

Specialised Connective Tissue

Cartilage, bones and blood are various types of **specialised connective tissues**. (The chapter **summary** classifies **cartilage, bone, blood, and adipose tissue** together as specialised connective tissues; section 7.1.2 above first classifies adipose as a loose connective tissue. Both source statements are preserved as written.) **Cartilage and bone are both structural materials**.

- **Cartilage** - its intercellular material is **solid and pliable** and **resists compression**. Cells of this tissue (**chondrocytes**) are enclosed in **small cavities within the matrix** secreted by them (Figure 7.6a).
 - Most of the cartilages in **vertebrate embryos** are **replaced by bones in adults**. In adults, cartilage is present in the **tip of nose, outer ear joints, between adjacent bones of the vertebral column, limbs and hands**.
- **Bone** - has a **hard and non-pliable ground substance rich in calcium salts and collagen fibres** which give bone its strength (Figure 7.6b). It is the **main tissue that provides structural frame to the body**.
 - Bones **support and protect softer tissues and organs**. The bone cells (**osteocytes**) are present in the spaces called **lacunae**.
 - **Limb bones**, such as the long bones of the legs, serve **weight-bearing functions**. They also interact with **skeletal muscles** attached to them to bring about movements. The **bone marrow** in some bones is the **site of production of blood cells**.
- **Blood** - a **fluid connective tissue** containing **plasma, red blood cells (RBC), white blood cells (WBC) and platelets** (Figure 7.6c). It is the main **circulating fluid** that helps in the **transport of various substances**.

You will learn more about blood in **Chapters 17 and 18**.

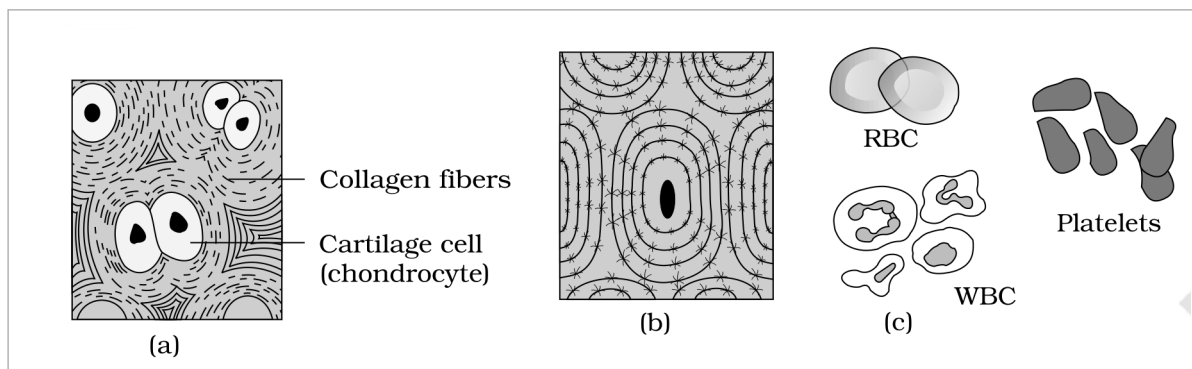


Figure 7.6 Specialised connective tissues: (a) Cartilage (b) Bone (c) Blood

① [NOTE] Figure 7.6 labels (verbatim): **Collagen fibers; Cartilage cell (chondrocyte); RBC; WBC; Platelets.**

7.1.3 Muscle Tissue

Each muscle is made of many **long, cylindrical fibres** arranged in **parallel arrays**. These fibres are composed of numerous fine fibrils, called **myofibrils**. Muscle fibres **contract (shorten)** in response to stimulation, then **relax (lengthen)** and return to their uncontracted state in a coordinated fashion. Their action moves the body to **adjust to changes in the environment** and to **maintain the positions of the various parts of the body**. In general, muscles play an **active role in all the movements of the body**.

Muscles are of **three types: skeletal, smooth, and cardiac** (Figure 7.7).

Muscle	Striations	Location	Control / junctions
Skeletal	Striated (striped)	Closely attached to skeletal bones; e.g., biceps - fibres bundled in a parallel fashion, several bundles enclosed by a sheath of tough connective tissue	Voluntary (learn more in Chapter 20)
Smooth	No striations; fibres taper at both ends (fusiform)	Wall of internal organs such as blood vessels, stomach and intestine	Involuntary - functioning cannot be directly controlled; cell junctions hold fibres together, bundled in a connective tissue sheath
Cardiac	Contractile tissue	Present only in the heart	Cell junctions fuse the plasma membranes; communication junctions (intercalated discs) let the cells contract as a unit

Cardiac muscle makes up the **contractile walls of the heart**: when one cell receives a signal to contract, its neighbours are also stimulated to contract.

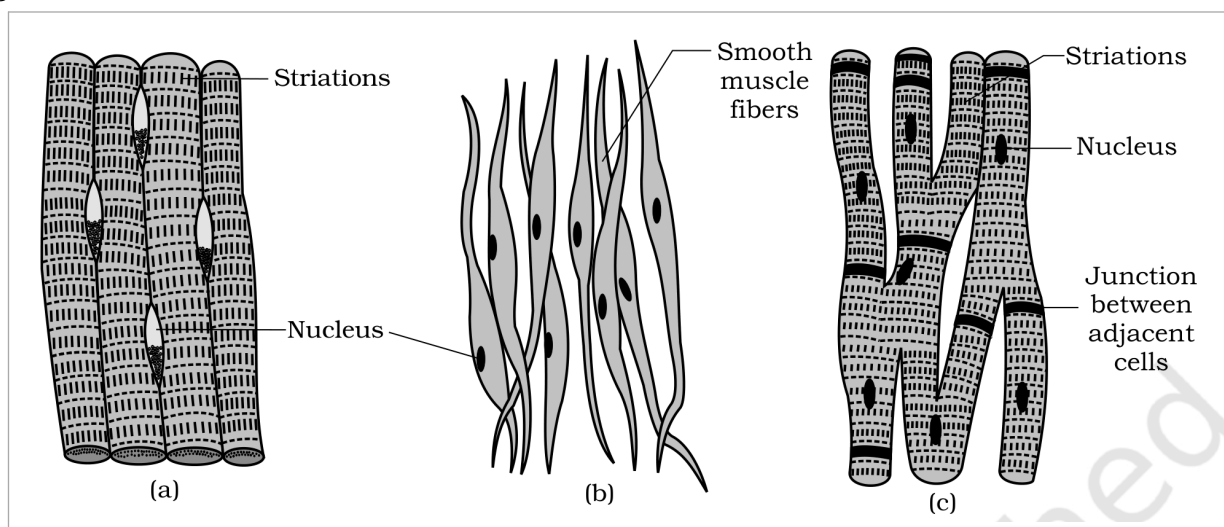


Figure 7.7 Muscle tissue: (a) Skeletal (striated) muscle tissue (b) Smooth muscle tissue (c) Cardiac muscle tissue

① [NOTE] Figure 7.7 labels (verbatim): **Nucleus; Striations; Smooth muscle fibers; Junction between adjacent cells.**

7.1.4 Neural Tissue

Neural tissue exerts the **greatest control** over the body's responsiveness to changing conditions. **Neurons**, the unit of the neural system, are **excitable cells** (Figure 7.8). The **neuroglial cells**, which constitute the rest of the neural system, **protect and support neurons**. Neuroglia make up **more than one-half the volume** of neural tissue in our body.

When a neuron is suitably stimulated, an **electrical disturbance** is generated which swiftly travels along its **plasma membrane**. Arrival of the disturbance at the neuron's endings, or **output zone**, triggers events that may cause **stimulation or inhibition** of adjacent neurons and other cells (details in **Chapter 21**).

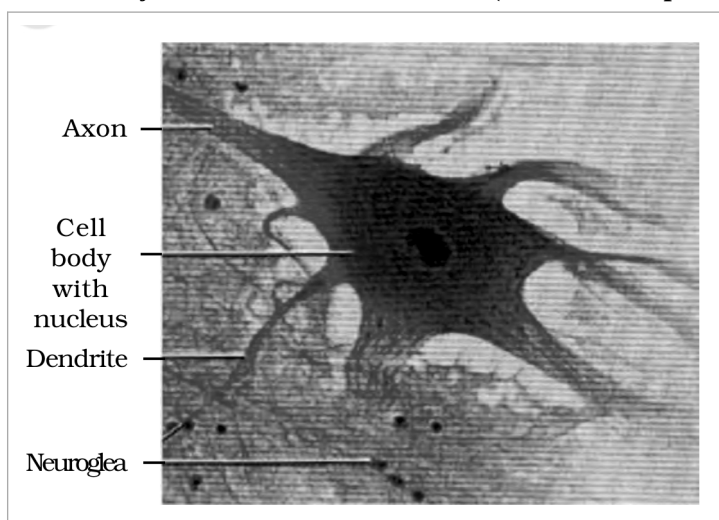


Figure 7.8 Neural tissue (Neuron with neuroglia)

① [NOTE] Figure 7.8 labels (verbatim): **Axon; Cell body with nucleus; Dendrite; Neuroglia.**

7.2 Organ and Organ System

The basic tissues mentioned above **organise to form organs** which in turn associate to form **organ systems** in the multicellular organisms. Such an organisation is essential for **more efficient and better coordinated activities** of millions of cells constituting an organism. **Each organ** in our body is made of **one or more type of tissues**. For example, our **heart** consists of **all the four types of tissues** - epithelial, connective, muscular and neural.

We also notice, after some careful study, that the complexity in organ and organ systems displays a certain discernable trend. This is called the **evolutionary trend** (details in class XII). You are being introduced to the **morphology and anatomy of three organisms at different evolutionary levels** to show their organisation and functioning.

- **Morphology** - the study of **form or externally visible features**. In plants or microbes, the term morphology precisely means only this; in animals it refers to the **external appearance** of the organs or parts of the body.
- **Anatomy** - conventionally used for the study of the **morphology of internal organs** in animals.

① [NOTE] In the requested NEET syllabus, the **Earthworm** (section 7.3) has been removed, so the chapter continues from the general organ/organ-system idea directly to the **Cockroach** and the **Frog** as the worked examples of animal organisation.

7.4 Cockroach

Cockroaches are **brown or black bodied** animals that are included in class **Insecta** of Phylum **Arthropoda**. Bright **yellow, red and green** coloured cockroaches have also been reported in tropical regions.

- Their **size** ranges from **1/4 inches to 3 inches (0.6-7.6 cm)** and they have **long antennae, legs** and a **flat extension of the upper body wall that conceals the head**.

- They are **nocturnal omnivores** that live in **damp places** throughout the world. They have become residents of human homes and thus are **serious pests and vectors of several diseases**.

7.4.1

Morphology

The adults of the common species of cockroach, *Periplaneta americana*, are about **34-53 mm long** with **wings that extend beyond the tip of the abdomen in males**. The body of the cockroach is **segmented** and divisible into **three distinct regions - head, thorax and abdomen** (Figure 7.14).

The entire body is covered by a **hard chitinous exoskeleton** (brown in colour). In each segment, the exoskeleton has hardened plates called **sclerites** (**tergites** dorsally and **sternites** ventrally) that are joined to each other by a **thin and flexible articular membrane (arthrodial membrane)**.

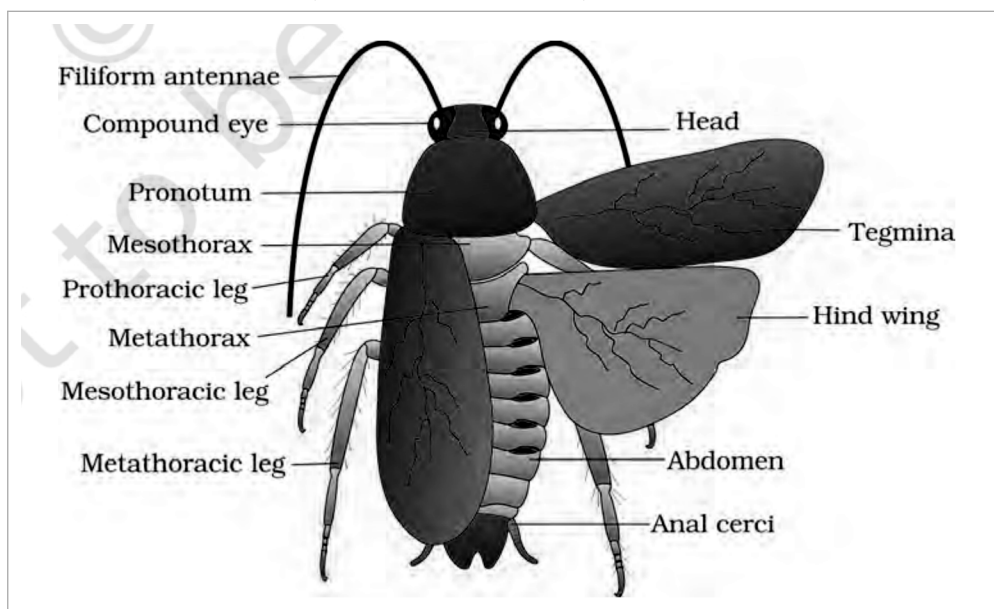


Figure 7.14 External features of cockroach

① [NOTE] Figure 7.14 labels (verbatim): **Filiform antennae; Compound eye; Head; Pronotum; Tegmina; Hind wing; Mesothorax; Prothoracic leg; Metathorax; Mesothoracic leg; Metathoracic leg; Abdomen; Anal cerci.**

Head

Head

The **head** is **triangular in shape** and lies anteriorly at **right angles to the longitudinal body axis**. It is formed by the **fusion of six segments** and shows **great mobility in all directions** due to a flexible neck (Figure 7.15).

- The head capsule bears a **pair of compound eyes**. A pair of **thread-like antennae** arise from **membranous sockets** lying in front of the eyes; antennae have **sensory receptors** that help in **monitoring the environment**.
- The anterior end of the head bears appendages forming **biting and chewing type of mouthparts**, consisting of a **labrum (upper lip)**, a pair of **mandibles**, a pair of **maxillae** and a **labium (lower lip)**. A median flexible lobe, acting as a tongue (**hypopharynx**), lies within the cavity enclosed by the mouthparts (Figure 7.15b).

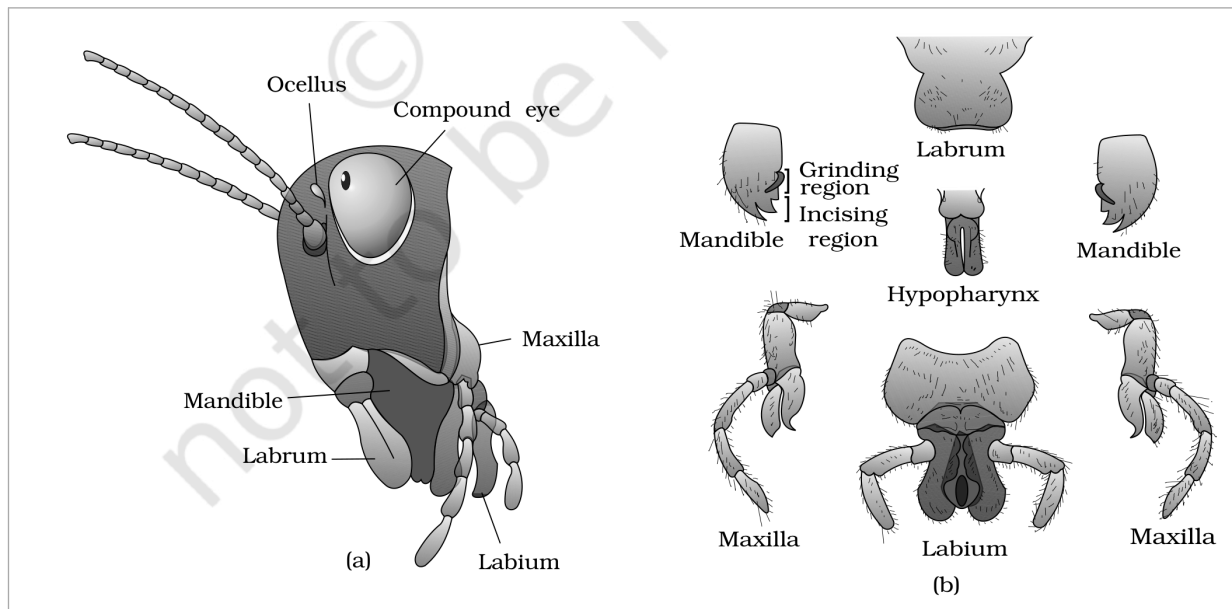


Figure 7.15 Head region of cockroach: (a) parts of head region (b) mouth parts

① [NOTE] Figure 7.15 labels (verbatim): **Ocellus; Compound eye; Mandible; Maxilla; Labrum; Labium; Grinding region; Incising region; Hypopharynx.**

Thorax

The **thorax** consists of **three parts - prothorax, mesothorax and metathorax**. The head is connected with the thorax by a short extension of the prothorax known as the **neck**. Each **thoracic segment bears a pair of walking legs**.

- The **first pair of wings** arises from the **mesothorax** and the **second pair** from the **metathorax**. (Thus **two pairs of wings** are present, **one pair each on the 2nd and 3rd thoracic segment**.)
- **Forewings (mesothoracic)** called **tegmina** are **opaque, dark and leathery** and cover the hind wings when at rest. The **hind wings** are **transparent, membranous** and are used in **flight**.

Abdomen

The **abdomen** in both males and females consists of **10 segments**.

- In **females**, the **7th sternum** is boat-shaped and together with the **8th and 9th sterna** forms a **brood or genital pouch** whose anterior part contains the **female gonopore, spermathecal pores and collateral glands**.
- In **males**, the **genital pouch or chamber** lies at the hind end of the abdomen, bounded dorsally by the **9th and 10th terga** and ventrally by the **9th sternum**. It contains the **dorsal anus, ventral male genital pore and gonapophysis**.
- Males bear a pair of short, thread-like **anal styles** which are **absent in females**. In both sexes, the **10th segment** bears a pair of jointed filamentous structures called **anal cerci**.

7.4.2 Anatomy

Dig Digestive System

The **alimentary canal** present in the body cavity is divided into **three regions: foregut, midgut and hindgut** (Figure 7.16). A **pair of salivary glands** is present near the crop.

- 1 The **mouth** opens into a short tubular **pharynx**, leading to a narrow tubular passage called the **oesophagus**.
- 2 The oesophagus opens into a sac-like structure called the **crop**, used for **storing food**.
- 3 The crop is followed by the **gizzard or proventriculus**: it has an outer layer of thick circular muscles and a thick inner cuticle forming **six highly chitinous plates called teeth**, and it **helps in grinding the food particles**. The entire foregut is lined by cuticle.

4

A ring of **6-8 blind tubules** called **hepatic or gastric caeca** is present at the **junction of foregut and midgut**, which **secrete digestive juice**.

5

At the **junction of midgut and hindgut** is present another ring of **100-150 yellow coloured thin filamentous Malpighian tubules**, which **help in removal of excretory products from haemolymph**.

6

The **hindgut** is broader than the midgut and is differentiated into **ileum, colon and rectum**; the **rectum opens out through the anus**.

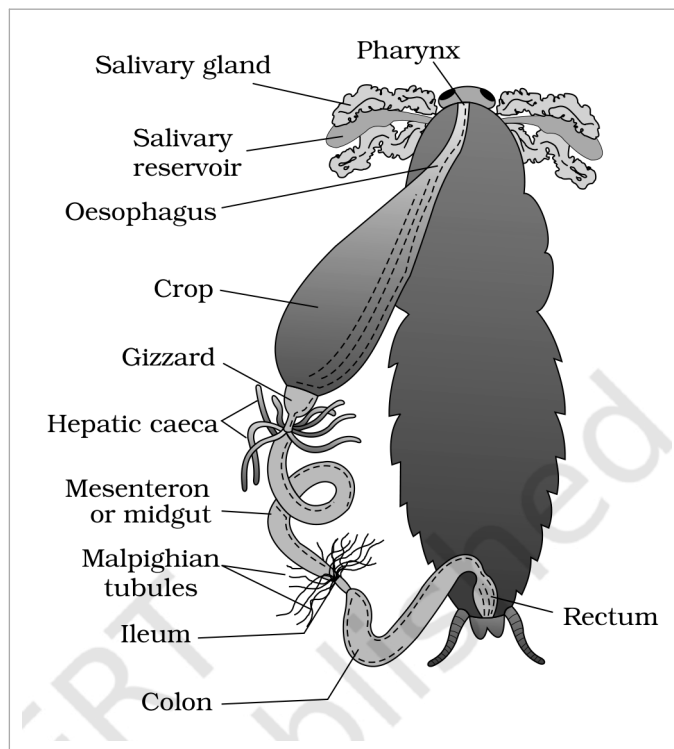


Figure 7.16 Alimentary canal of cockroach

① [NOTE] Figure 7.16 labels (verbatim): **Salivary gland; Pharynx; Salivary reservoir; Oesophagus; Crop; Gizzard; Hepatic caeca; Mesenteron or midgut; Malpighian tubules; Ileum; Colon; Rectum.**

Circ

Circulatory System

The **blood vascular system** of cockroach is an **open type** (Figure 7.17). Blood vessels are **poorly developed** and open into a space (**haemocoel**). Visceral organs located in the haemocoel are **bathed in blood (haemolymph)**. The haemolymph is composed of **colourless plasma and haemocytes**.

The **heart** of cockroach consists of an **elongated muscular tube** lying along the **mid-dorsal line** of thorax and abdomen. It is differentiated into **funnel-shaped chambers with ostia** on either side. **Blood from sinuses enters the heart through ostia** and is **pumped anteriorly to the sinuses again**.

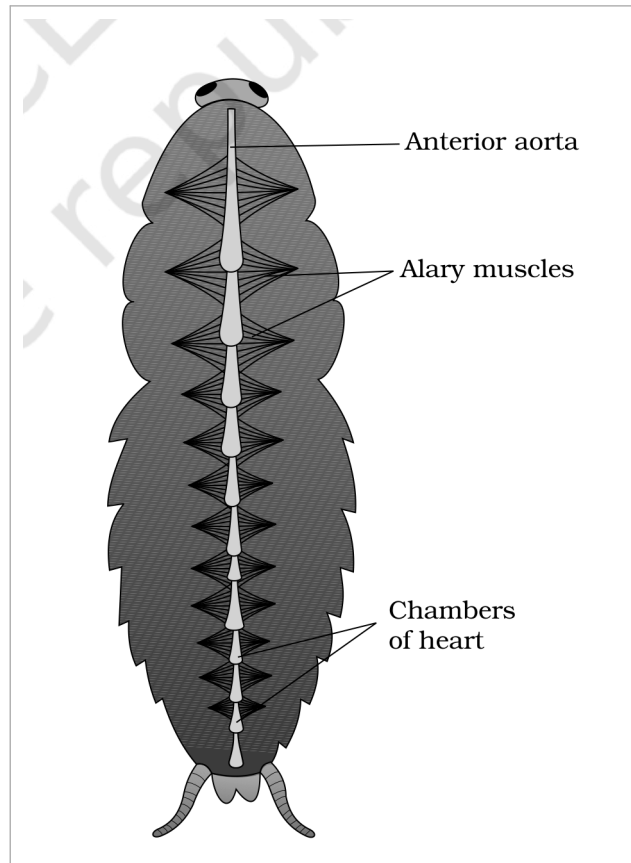


Figure 7.17 Open circulatory system of cockroach

① [NOTE] Figure 7.17 labels (verbatim): **Anterior aorta**; **Alary muscles**; **Chambers of heart**.

Resp

Respiratory System

The **respiratory system** consists of a **network of trachea**, that open through **10 pairs of small holes called spiracles** present on the lateral side of the body. **Thin branching tubes** (tracheal tubes subdivided into **tracheoles**) carry **oxygen from the air to all the parts**. The opening of the spiracles is regulated by the **sphincters**. **Exchange of gases** takes place at the tracheoles by **diffusion**.

Excr

Excretory System

Excretion is performed by **Malpighian tubules**. Each tubule is lined by **glandular and ciliated cells**. They **absorb nitrogenous waste products** and **convert them into uric acid** which is excreted out through the hindgut. Therefore, this insect is called **uricotelic**. In addition, the **fat body**, **nephrocytes** and **urecose glands** also help in excretion.

Nerv

Nervous System

The **nervous system** of cockroach consists of a series of **fused, segmentally arranged ganglia** joined by **paired longitudinal connectives** on the ventral side. **Three ganglia lie in the thorax, and six in the abdomen**. The nervous system is **spread throughout the body**: the head holds a bit of it while the rest is situated along the **ventral (belly-side) part** of the body.

① [NOTE] Because the nervous system is spread out, if the head of a cockroach is cut off it will still **live for as long as one week**.

In the head region, the **brain** is represented by the **supra-oesophageal ganglion** which supplies nerves to the **antennae and compound eyes**. In cockroach, the **sense organs** are **antennae, eyes, maxillary palps, labial palps, anal cerci**, etc.

The **compound eyes** are situated at the dorsal surface of the head. Each eye consists of about **2000 hexagonal ommatidia** (sing.: ommatidium). With the help of several ommatidia, a cockroach can receive **several images of an object**. This kind of vision is known as **mosaic vision**, with **more sensitivity but less resolution**, being common during night (hence called **nocturnal vision**).

Cockroaches are **dioecious** and both sexes have **well developed reproductive organs** (Figure 7.18).

- The **male reproductive system** consists of a **pair of testes**, one lying on each lateral side in the **4th-6th abdominal segments**. From each testis arises a thin **vas deferens**, which opens into the **ejaculatory duct** through the **seminal vesicle**. The ejaculatory duct opens into the **male gonopore** situated ventral to the anus.
- A characteristic **mushroom-shaped gland** is present in the **6th-7th abdominal segments** which functions as an **accessory reproductive gland**. The external genitalia are represented by **male gonapophysis or phallomere** (chitinous asymmetrical structures surrounding the male gonopore).
- The **sperms** are stored in the seminal vesicles and are glued together in the form of bundles called **spermatophores** which are discharged during copulation.
- The **female reproductive system** consists of **two large ovaries**, lying laterally in the **2nd-6th abdominal segments**. Each ovary is formed of a group of **eight ovarian tubules or ovarioles**, containing a chain of developing ova. **Oviducts** of each ovary unite into a single **median oviduct (also called vagina)** which opens into the genital chamber.
- A pair of **spermatheca** is present in the **6th segment** which opens into the genital chamber. **Sperms are transferred through spermatophores**.
- Their **fertilised eggs** are encased in capsules called **oothecae**. An ootheca is a **dark reddish to blackish brown capsule, about 3/8 inch (8 mm) long**. They are dropped or glued to a suitable surface, usually in a crack or crevice of **high relative humidity near a food source**. On an average, females produce **9-10 oothecae, each containing 14-16 eggs**.
- The development of **P. americana** is **paurometabolous**, meaning there is development through a **nymphal stage**. The **nymphs look very much like adults**. The nymph grows by **moulting about 13 times** to reach the adult form. The **next to last nymphal stage has wing pads** but only adult cockroaches have wings. (After the rupturing of a single ootheca, **sixteen young ones, called nymphs**, come out.)

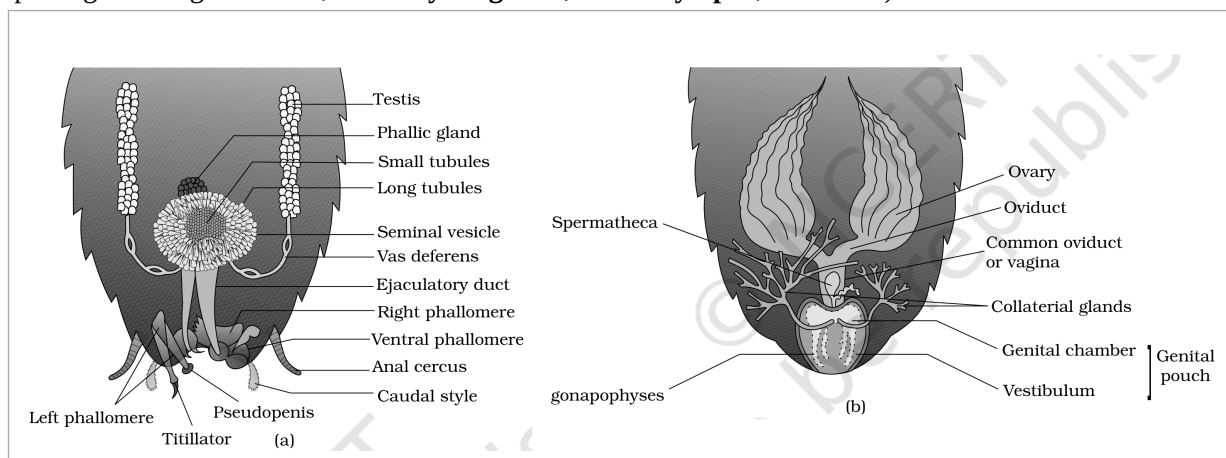


Figure 7.18 Reproductive system of cockroach: (a) male (b) female

① [NOTE] Figure 7.18 labels (verbatim): **Testis; Phallic gland; Small tubules; Long tubules; Seminal vesicle; Vas deferens; Ejaculatory duct; Right phallomere; Ventral phallomere; Anal cercus; Caudal style; Pseudopenis; Titillator; Left phallomere; Ovary; Oviduct; Common oviduct or vagina; Collateral glands; Genital chamber; Vestibulum; Genital pouch; Spermatheca; Gonapophyses.**

Many species of cockroaches are **wild** and are of **no known economic importance** yet. A few species **thrive in and around human habitat**. They are **pests** because they **spoil food and contaminate it with their smelly excreta**, and they can **transmit a variety of bacterial diseases** by contaminating food material.

Frogs can live both on **land and in freshwater** and belong to class **Amphibia** of phylum **Chordata**. The most common species of frog found in India is **Rana tigrina**. They **do not have a constant body temperature** - their body temperature varies with the temperature of the environment; such animals are called **cold blooded or poikilotherms**.

- You might have noticed **changes in the colour** of the frogs while they are in grasses and on dry land. They have the ability to **change the colour to hide them from their enemies (camouflage)**. This **protective coloration is called mimicry**.
- Frogs are **not seen during peak summer and winter**. During this period they take shelter in **deep burrows** to protect them from extreme heat and cold. This is known as **summer sleep (aestivation)** and **winter sleep (hibernation)** respectively.

The **skin** of a frog is **smooth and slippery** due to the presence of **mucus** and is **always maintained in a moist condition**. **Mucous glands** are present in the skin, which is **highly vascularised and helps in respiration** both in water and on land. The colour of the **dorsal side** of the body is generally **olive green with dark irregular spots**, while on the **ventral side** the skin is **uniformly pale yellow**. The frog **never drinks water but absorbs it through the skin**.

The body of a frog is divisible into **head and trunk** (Figure 7.19); a **neck and tail are absent**.

- Above the mouth, a **pair of nostrils** is present. **Eyes** are bulged and covered by a **nictitating membrane** that protects them while in water. On either side of the eyes a **membranous tympanum (ear)** receives sound signals.
- The **forelimbs and hind limbs** help in **swimming, walking, leaping and burrowing**. The **hind limbs end in five digits** and are **larger and muscular** than the **fore limbs that end in four digits**. **Feet have webbed digits** that help in swimming.
- Frogs exhibit **sexual dimorphism**. **Male frogs** can be distinguished by the presence of **sound producing vocal sacs** and also a **copulatory pad on the first digit of the fore limbs**, which are **absent in female frogs**.

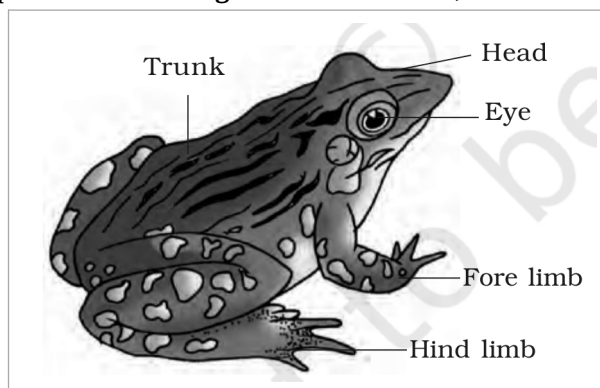


Figure 7.19 External features of frog

① **[NOTE]** Figure 7.19 labels (verbatim): **Eye; Fore limb; Hind limb; Head; Trunk**.

The **body cavity** of frogs accommodates different organ systems - **digestive, circulatory, respiratory, nervous, excretory and reproductive** - with well developed structures and functions (Figure 7.20).

The **digestive system** consists of the **alimentary canal** and **digestive glands**. The alimentary canal is **short** because frogs are **carnivores** and hence the length of the intestine is reduced. **Food is captured by the bilobed tongue**.

The **mouth** opens into the **buccal cavity** that leads to the **oesophagus** through the **pharynx**.

2

The oesophagus is a short tube that opens into the **stomach**. Digestion of food takes place by the action of **HCl and gastric juices** secreted from the walls of the stomach.

3

Partially digested food called **chyme** passes from the stomach to the first part of the small intestine, the **duodenum**, which receives **bile from the gall bladder** and **pancreatic juices from the pancreas** through a **common bile duct**.

4

Bile emulsifies fat and **pancreatic juices digest carbohydrates and proteins**; **final digestion** takes place in the intestine.

5

Digested food is absorbed by the numerous finger-like folds in the inner wall of the intestine called **villi and microvilli**.

6

The **undigested solid waste** moves into the **rectum** and passes out through the **cloaca**.

The main **digestive glands** are the **liver**, which secretes **bile** that is stored in the **gall bladder**, and the **pancreas**, which produces **pancreatic juice** containing digestive enzymes.

Resp

Respiratory System

Frogs **respire on land and in the water by two different methods**. In water, the **skin acts as the aquatic respiratory organ (cutaneous respiration)** - **dissolved oxygen** in the water is exchanged through the skin by **diffusion**. On land, the **buccal cavity, skin and lungs** act as the respiratory organs; respiration by lungs is called **pulmonary respiration**.

- The **lungs** are a pair of **elongated, pink coloured sac-like structures** present in the upper part of the trunk region (thorax). **Air enters through the nostrils** into the buccal cavity and then to the lungs. During **aestivation and hibernation**, gaseous exchange takes place through the skin.

Circ

Circulatory System

The **vascular system** of frog is a **well-developed closed type**. Frogs also have a **lymphatic system**. The **blood vascular system** involves **heart, blood vessels and blood**; the **lymphatic system** consists of **lymph, lymph channels and lymph nodes**. The circulatory system is **closed with single circulation**.

- The **heart** is a muscular structure situated in the upper part of the body cavity. It has **three chambers - two atria and one ventricle** - and is covered by a membrane called **pericardium**. A triangular structure called the **sinus venosus** joins the **right atrium** and receives blood through the major veins called the **vena cava**. The **ventricle** opens into a sac-like **conus arteriosus** on the ventral side of the heart.
- The blood from the heart is carried to all parts of the body by the **arteries (arterial system)**. The **veins** collect blood from different parts of the body to the heart and form the **venous system**. Special venous connections are present: between **liver and intestine** (the **hepatic portal system**) and between the **kidney and lower parts of the body** (the **renal portal system**).
- The **blood** is composed of **plasma and cells**. The blood cells are **RBC (red blood cells) or erythrocytes, WBC (white blood cells) or leucocytes and platelets**. **RBCs are nucleated** and contain the red coloured pigment **haemoglobin**.
- The **lymph** is different from blood - it **lacks few proteins and RBCs**. The blood **carries nutrients, gases and water** to the respective sites during circulation, which is achieved by the **pumping action of the muscular heart**.

Excr

Excretory System

The **elimination of nitrogenous wastes** is carried out by a **well developed excretory system** consisting of a pair of **kidneys, ureters, cloaca and urinary bladder**. The kidneys are **compact, dark red and bean-like structures** situated a little posteriorly in the body cavity on both sides of the vertebral column. Each kidney is composed of several structural and functional units called **uriniferous tubules or nephrons**.

- **Two ureters** emerge from the kidneys in **male frogs**; the ureters act as a **urinogenital duct** which opens into the cloaca. In **females**, the **ureters and oviduct open separately** in the cloaca. The **thin-walled urinary bladder** is present ventral to the rectum and also opens into the cloaca.

- The frog **excretes urea** and thus is a **ureotelic animal**. Excretory wastes are **carried by blood into the kidney** where they are **separated and excreted**.

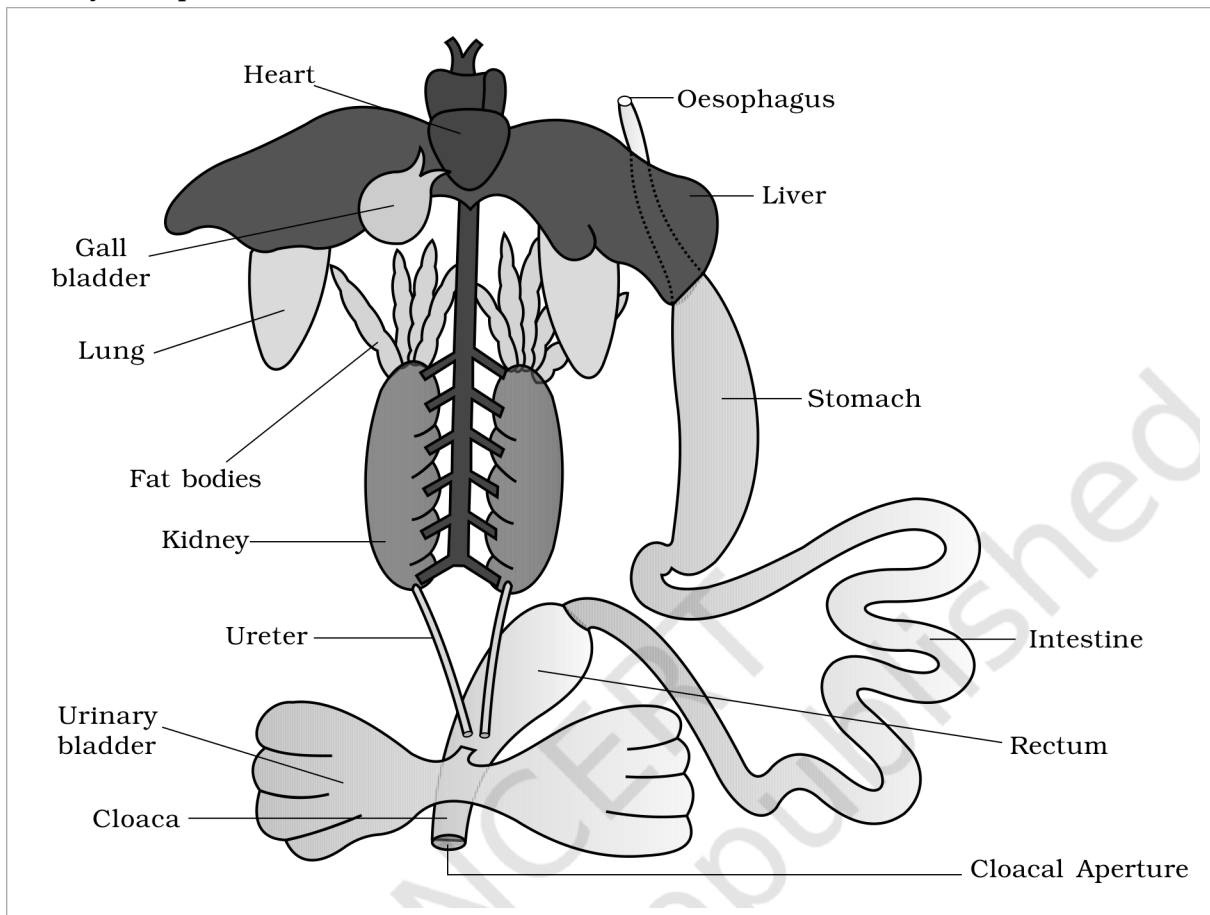


Figure 7.20 Diagrammatic representation of internal organs of frog showing complete digestive system

① [NOTE] Figure 7.20 labels (verbatim): **Heart; Oesophagus; Liver; Gall bladder; Lung; Fat bodies; Kidney; Ureter; Urinary bladder; Cloaca; Stomach; Intestine; Rectum; Cloacal Aperture.**

Ctrl

Control and Coordination

The system for **control and coordination** is **highly evolved** in the frog and includes both the **neural system and endocrine glands**. The **chemical coordination** of various organs is achieved by **hormones** secreted by the **endocrine glands**. The prominent endocrine glands found in the frog are the **pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals and gonads**.

- The **nervous system** is organised into a **central nervous system (brain and spinal cord)**, a **peripheral nervous system (cranial and spinal nerves)** and an **autonomic nervous system (sympathetic and parasympathetic)**. There are **ten pairs of cranial nerves** arising from the brain.
- The **brain** is enclosed in a bony structure called the **brain box (cranium)** and is divided into **fore-brain, mid-brain and hind-brain**. The **forebrain** includes **olfactory lobes, paired cerebral hemispheres and unpaired diencephalon**; the **midbrain** is characterised by a **pair of optic lobes**; the **hind-brain** consists of **cerebellum and medulla oblongata**. The medulla oblongata passes out through the **foramen magnum** and continues into the **spinal cord**, which is enclosed in the vertebral column.
- The frog has different types of **sense organs**: organs of **touch (sensory papillae)**, **taste (taste buds)**, **smell (nasal epithelium)**, **vision (eyes)** and **hearing (tympanum with internal ears)**. Of these, the **eyes and internal ears** are well-organised structures and the rest are **cellular aggregations around nerve endings**.
- **Eyes** in a frog are a pair of **spherical structures** situated in the orbit in the skull; these are **simple eyes (possessing only one unit)**. **External ear is absent** in frogs and only the **tympanum** can be seen externally. The **ear** is an organ of **hearing as well as balancing (equilibrium)**.

Frogs have **well organised male and female reproductive systems**.

- **Male reproductive organs** consist of a pair of **yellowish ovoid testes** (Figure 7.21), found adhered to the upper part of the kidneys by a double fold of peritoneum called the **mesorchium**. **Vasa efferentia** are **10-12 in number** and arise from the testes; they enter the kidneys on their side and open into **Bidder's canal**. Finally it communicates with the **urinogenital duct** that comes out of the kidneys and opens into the **cloaca**.
- The **cloaca** is a small, median chamber used to pass **faecal matter, urine and sperms** to the exterior.
- The **female reproductive organs** include a pair of **ovaries** (Figure 7.22), situated near the kidneys with **no functional connection** with them. A pair of **oviducts** arising from the ovaries opens into the **cloaca separately**. A mature female can lay **2500 to 3000 ova at a time**.
- **Fertilisation is external** and takes place in **water**. Development involves a larval stage called the **tadpole**, which undergoes **metamorphosis** to form the adult.

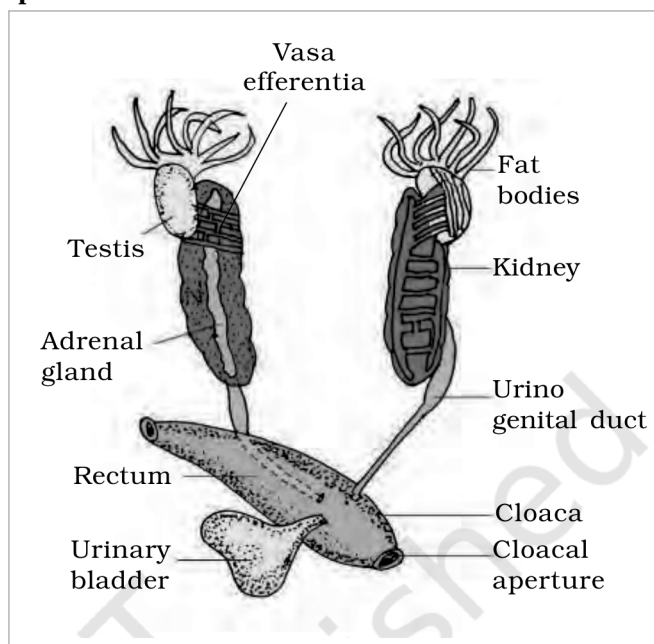


Figure 7.21 Male reproductive system

① [NOTE] Figure 7.21 labels (verbatim): **Vasa efferentia; Testis; Fat bodies; Kidney; Adrenal gland; Urinogenital duct; Rectum; Urinary bladder; Cloaca; Cloacal aperture.**

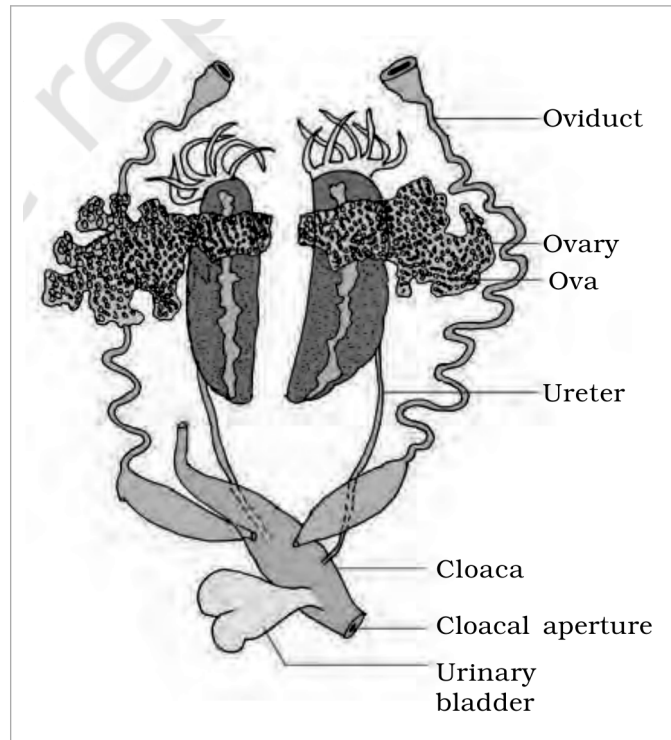


Figure 7.22 Female reproductive system

① [NOTE] Figure 7.22 labels (verbatim): **Oviduct; Ovary; Ova; Ureter; Cloaca; Cloacal aperture; Urinary bladder.**

Frogs are beneficial for mankind because they **eat insects and protect the crop**. They **maintain ecological balance** as an important link of the **food chain and food web** in the ecosystem. In some countries the **muscular legs of frog** are used as food by man.

Recap

Quick Recap

- A **tissue** is a group of cells along with intercellular substances performing **one or more functions**. Cells, tissues, organs and organ systems exhibit **division of labour** for the survival of the whole body. Complex animals have **four tissue types**: epithelial, connective, muscular and neural.
- **Epithelia** are sheet-like tissues lining the body's surface and its **cavities, ducts and tubes**, with one **free surface**; their cells are connected at **tight, adhering and gap junctions**. Simple epithelium is squamous, cuboidal or columnar; compound epithelium is protective.
- **Connective tissues** bind together, support, strengthen, protect and insulate other tissues, and cover all three other tissue types. **Soft connective tissues** have protein fibres and varied cells in a ground substance. **Cartilage, bone, blood and adipose tissue** are specialised connective tissues; cartilage and bone are structural materials, **blood is a fluid transport tissue**, and **adipose tissue is a reservoir of stored energy**.
- **Muscle tissue** contracts in response to stimulation and helps movement. **Skeletal muscle** is attached to bones, **smooth muscle** is a component of internal organs, and **cardiac muscle** makes up the contractile walls of the heart. **Nervous tissue** exerts the greatest control over the body's responses; **neurons** are its basic units.
- The body of **Cockroach (*Periplaneta americana*)** is covered by a chitinous exoskeleton and divided into **head, thorax and abdomen**, whose segments bear jointed appendages. The **three thoracic segments** each bear a pair of walking legs, **two pairs of wings** are present (one pair each on the 2nd and 3rd segment), and the **abdomen has 10 segments**.
- The cockroach **alimentary canal** has a mouth with mouthparts, pharynx, oesophagus, crop, gizzard, midgut, hindgut and anus, with a **pair of salivary glands near the crop**. **Hepatic caecae** lie at the foregut-midgut junction and **Malpighian tubules** at the midgut-hindgut junction (excretion). The blood vascular system is **open**; respiration is by a network of **tracheae** opening through **spiracles**; the nervous system is **segmentally arranged ganglia with a ventral nerve cord**.
- In the cockroach, a **pair of testes** is present in the 4th-6th segments and **ovaries** in the 2nd-6th segments; **fertilisation is internal**. A female produces **9-10 oothecae** bearing developing embryos, and after rupturing of a

single ootheca **sixteen nymphs** come out.

- The **Indian bullfrog, Rana tigrina**, is covered by **skin** with mucous glands (highly vascularised, aids respiration in water and on land), and the body is divisible into **head and trunk**. A muscular **bilobed tongue** captures prey. Its alimentary canal (oesophagus, stomach, intestine, rectum) opens into the **cloaca**; the main **digestive glands are liver and pancreas**.
- The frog can respire through **skin in water and lungs on land**. Its circulatory system is **closed with single circulation** and **RBCs are nucleated**. The nervous system is organised into **central, peripheral and autonomic** parts. The urinogenital system has **kidneys and urinogenital ducts** opening into the cloaca.
- The male reproductive organ is a **pair of testes** and the female a **pair of ovaries**. A female lays **2500-3000 ova** at a time; **fertilisation and development are external**, and the eggs hatch into **tadpoles** which metamorphose into frogs.

Appendix

Terms Used in the Exercises

A few exercise questions use terms that the supplied chapter text does not itself define. Those **gap** terms are answered here so the exercises can be attempted from these notes alone; every other exercise is answerable from the sections and figures above.

- **Axon** [Q7(b)] - the **single long process of a neuron** that conducts the nerve impulse **away from the cell body**, towards the neuron's endings (output zone). It is labelled in Figure 7.8 but not defined in the running text.
- **Simple gland vs compound gland** [Q9(e)] - a **simple gland** has an **unbranched duct**, whereas a **compound gland** has a **branched duct system** draining many secretory units. (This is distinct from the unicellular/multicellular classification given in 7.1.1.)
- **Coxa** [Q10(e)] - the **basal segment of an insect leg**, which attaches the leg to the thoracic segment. (The cockroach's thoracic segments - mesothorax and metathorax - are described in 7.4.1.)
- **Protonema** [Q10(e)] - the **green, branched, filamentous first stage** of the gametophyte in the life cycle of a **moss**. It is a plant structure and does not belong to the animals studied in this chapter.
- **Typhlosole** [Q11(e)] - the **internal median fold of the dorsal wall of the intestine of the earthworm**, which **increases the absorptive surface area**. (Earthworm is outside the NEET syllabus; the term is listed only because an exercise assumes it.)